

Breathing New Energy

A Garment Offering Comfort and Haptic Breathing Guidance to Regain Breath Control and Energy

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Abstract

Long COVID is a condition that affects around 6% of those who have been infected with the coronavirus. It causes persistent symptoms such as fatigue, struggle with concentration, hypersensitivity and breathlessness. Due to its recent prevalence, few support tools have been developed to address this condition. *Breathing New Energy* presents a garment designed to offer comfort, biofeedback and haptic breathing guidance through on the belly, promoting breath control and relaxation. Reflective Transformative Design Process (RTDP) was used to structure the process, containing cultural probes with the user group, user interviews, expert insights, and iterative prototyping along with textiles and electronics. Results show that individuals with long COVID prefer warmth, softness and a loose fit in a comforting garment. Live LEDs provide biofeedback to create additional awareness of the body. Haptic breathing guidance through vibration on the belly can be an effective method to encourage relaxation and recharge energy. Thusly, *Breathing New Energy* looks to contribute to the field of wearable technology, embodied interaction, and inclusive healthcare design.

Ccs Concepts

• Human-centered computing~Interaction design • Applied computing~Life and medical sciences • Hardware~Communication hardware, interfaces and storage

Keywords

Wearable technology; Long COVID; Biofeedback; Breathing guidance

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1 Introduction

While the COVID-19 pandemic may feel like a distant memory to many, its long-term effects are still impacting people's lives today. According to the World Health Organization (WHO), approximately 6% of those who have been infected with the coronavirus experience symptoms that persist for more than three months, and in some cases, for years [25]. This condition is known as long COVID. Symptoms vary, but commonly include fatigue, struggle with concentration, hypersensitivity, shortness of breath, and Post-Exertional Malaise (PEM) [1, 2, 21]. Many symptoms are shared across other infection-associated chronic conditions, also called post-acute infection syndromes [1]. These physical limitations often lead to people being unable to work, exercise and socialize, causing higher rates of mental health disorders [2, 4]. Due to long COVID's short existence and complex nature there are currently very few support tools specifically designed to meet the needs of individuals with long COVID. Motivated by an interest in inclusive, garment-based, and medical design, this project explores how wearables can support an underrecognized group, especially since long COVID affects twice as many women as men [20]. *Breathing New Energy*, explores how wearable technology can support the well-being of people with long COVID through direct contact with the body, requiring less active effort. The wearable technology aims to offer comfort using the soothing properties of textiles, while technology delivers biofeedback and haptic breathing guidance to create awareness of the breath and physical boundaries, encouraging relaxation to recharge energy.

2 Related Work

People with long COVID often require multidisciplinary care, and online forums currently support with shared patient experiences or toolkits for symptom-specific treatments [4, 5, 13]. However, these solutions require physical and cognitive effort to understand and apply. A wearable that offers direct feedback could help reduce this cognitive load and support users in a more embodied, intuitive way.

Studies show that slow abdominal breathing can reduce anxiety, lower blood pressure, and improve the functioning of the nervous system [18, 27]. According to Van Voorthuizen et al. [2022], a dysregulated breathing pattern could be causing the breathlessness often experienced in long COVID. Retraining abdominal breathing in people with Chronic Fatigue Syndrome

(CFS), which has similar symptoms as long COVID, has shown improvements in tidal volume and respiratory rate [11, 15]. Another study shows that breathing exercises have been linked to increased lung capacity and slower respiration rates in people with long COVID [10].

There are various techniques to measure respiratory rate with wearables. Examples include piezoelectric patches (RespiraSense [14]), chest bands with inductive plethysmography (Hexoskin [8]), and sensors that use sound or ECG-derived data [6, 17]. Piezoelectric sensors performed well even during movement [12]. However, consumer versions are hard to access, so *Breathing New Energy* explores using an accessible stretch sensor from Robotshop that changes resistance when stretched [16].

Biofeedback, which involves translating real-time physiological data into signals such as light, sound, or haptic cues, has shown promise in helping users gain greater awareness and control over their bodies [26]. Studies have shown that respiration-based biofeedback can help lower stress and improve breathing [23]. The following wearables present how breathing guidance can support relaxation and awareness without demanding constant attention, especially relevant for people with long COVID. The Wearable Breathing Trainer by Siering et al. [2019] uses light and haptic pressure to help children improve their breath rhythm, reducing cognitive load. The MysiShirt by TouchWaves BV uses vibration motors to guide breathing patterns for mindfulness [24].

3 Design Process

The overall design process, guided by the RTDP, described by Hummels et al. [2009], stimulates a dynamic and iterative way of designing activities and reflecting. RTDP encourages learning through making, seeking validation, and iterating.

3.1 User Study

A user study was conducted with five participants (4F, 1M; aged 21–65; 2 recovered, 3 with ongoing symptoms). User input was gained through online research, semi-structured interviews, and a cultural probe study described by Gaver et al [1999]. The cultural probe consisted of six postcards exploring users' energy management and comfort preferences. Participants responded via text or craft materials provided, generating inspirational qualitative input for the practical and structural design of the garment.

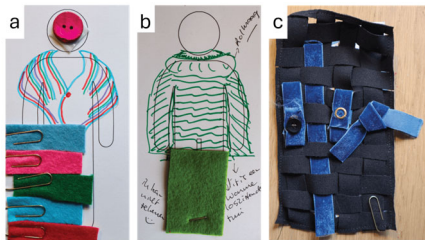


Figure 1: User input from cultural probe. a & b) Comforting garment described by two participants. c) Exploring tightening methods with the user.

Thematic Analysis (TA), based on Braun et al. [2006], was used to analyze the qualitative data from both interviews and probe results. From this TA, several design implications were derived: the garment should not be worn 24/7, rather practical than cognitive function (no mental logging), the garment should be soft, light, warm and loose, sound protection (like earplugs) could be included, the design should avoid over-stimulating the user.

3.2 Expert Perspective

Three medical experts in physiotherapy, breathing, and long COVID were interviewed. Their insights validated functional directions and emphasized that breathing guidance must be embedded with contextual sensitivity and technical accuracy.

3.3 Physicalizing

The design process explored various textiles and garments, with a scarf-like form considered early on for its comfort and wearability. As the focus shifted towards breathing guidance, various tightening and wrapping techniques were tested to measure breathing at both chest and abdominal levels. Tightening a knot from the same fabric was chosen for its adaptability, simplicity, and minimal sensory impact. Standard breathing sensor placement at chest height, as seen in Lanata et al. [2010], is suitable for male bodies but often uncomfortable or less accurate for female users. To address this, the chest sensor was adapted to sit just below the breasts, while still being able to distinguish between chest and abdominal breathing, making the design more inclusive for women who make up the majority of long COVID patients.

The electronics used were tested and integrated compactly in the final prototype (Figure 2). The components include: 2 x 15cm resistive stretch sensors, 3 x vibration motor modules, 1 x LED strip (20 LEDs), 2 x buttons, 1 x Seeeduino XIAO microcontroller, and 1 x 5V USB power bank. All electronic components are sewn into the textile using conductive yarn with 1Ω resistance/meter. Sensor accuracy was enhanced and validated in MATLAB. Using a Fast Fourier Transform (FFT) and bandpass filtering (0.1–0.8 Hz), the sensor readings can be isolated to relevant breathing frequencies. This allows the extraction of both respiratory rate and breathing depth from chest and abdomen sensors.

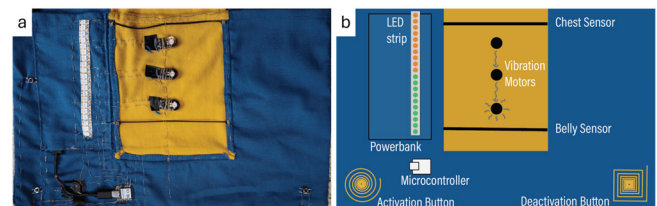


Figure 2: Electronics final prototype. a) The full circuit sewn into the inside of the garment. b) Graphical visual of the placement of the electrical components.

4 Design Implementation

Using the insights gained from the design process, a final design was physicalized in an interactive prototype called *Breathing New Energy*. This dual-modality garment can be worn either as a practical shirt or a comforting scarf (Figures 3 and 4). In both forms, it monitors the user's breathing using two stretch sensors placed on the chest and abdomen.

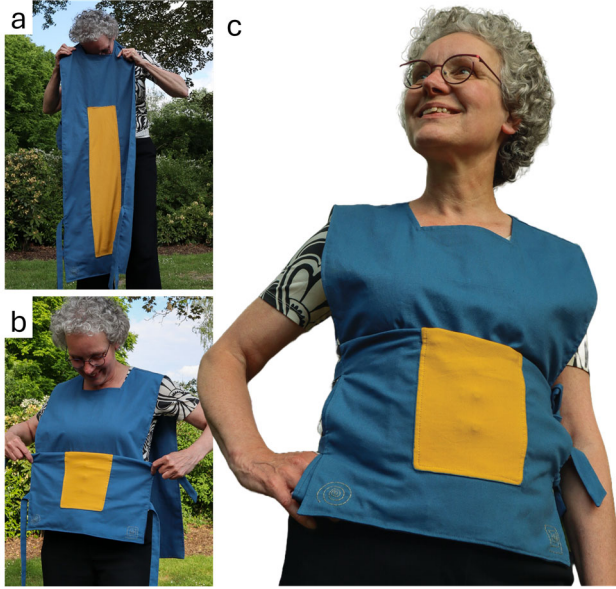


Figure 3: Shirt mode. a) Putting head through hole. b) Tying the garment around torso. c) The garment worn as a shirt.



Figure 4: Scarf mode. a) Tying the garment around the torso without the head through the hole. b) Draping the garment around the neck as desired. c) The garment worn as a scarf.

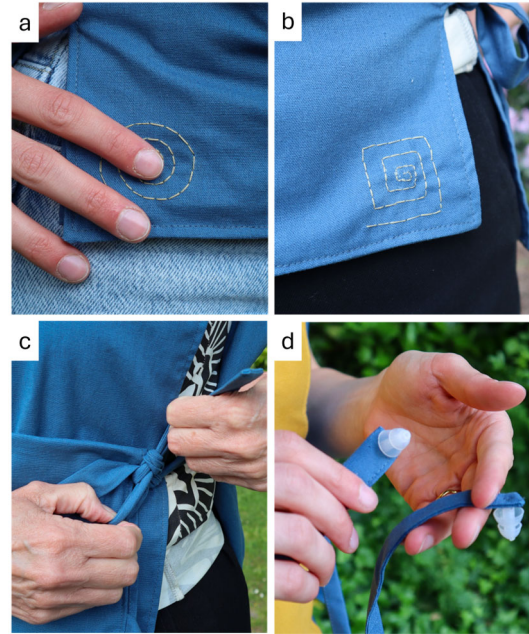


Figure 5: a) Activation button of breathing guidance being pressed. b) Deactivation button of breathing guidance. c) Tightening the garment with a knot. d) Earplugs attached to the garment.

When dysfunctional breathing is detected at rest (such as a high respiratory rate or dominant chest breathing), the garment starts a haptic breathing exercise. Gentle vibration moves feedforward from the chest to the belly, after which the bottom vibration motor increases and decreases in intensity with a pause in between, signaling the inhale and exhale rhythm the user should follow (Figure 6bcd). This feedforward vibration guides the user toward slower, abdominal breathing. Alternatively, the user can manually activate or deactivate the vibration guidance using buttons on the garment, maintaining control over when to engage (Figure 5ab).

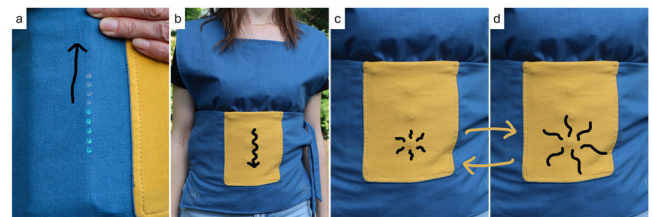


Figure 6: a) Biofeedback through real-time LED visualization of breathing. b) Vibration guidance moving from chest to belly. c & d) Increasing and decreasing guiding vibration intensity.

Soft LED lights visualize the breathing rhythm in real-time; increasing orange LEDs visualize chest inhalation, while increasing green LEDs show abdominal inhalation (Figure 6a). Showing this in real-time enhances awareness of the breath. In scarf mode, additional fabric and integrated earplugs offer a sense of protection from external auditory and extraneous visual stimuli (Figure 5d).

The garment's main functionalities include:

- **Creating awareness of one's breathing:** through LED feedback and vibration guidance from chest to belly.
- **Creating awareness of physical boundaries:** through the automatic start of the breathing guidance.
- **Guiding abdominal breathing exercises to promote relaxation:** by gradually slowing the breathing rhythm through vibration guidance.
- **Providing comfort by protecting auditory senses:** using earplugs and fabric that wraps the ears.
- **Offering non-verbal communication:** through physical gestures (e.g. covering eyes or ears) that can be made with the garment.

The materials were chosen with both function and comfort in mind: a non-stretch woven linen textile (blue) combined with a high stretch cotton spandex jersey (yellow) optimizes the stretch sensor's functionality. User options include linen for warmer climates or wool for cooler climate.

Though the vibration feedback is yet to be tested with long COVID users, healthy individuals on an interactive exhibition (Figure 7) described the feedback as calming, while some desired a stronger intensity. Future iterations could allow users to adjust vibration strength to personal needs.

This design aims to support long COVID users in creating awareness of own physical boundaries, guiding breath control and relaxation in order to recharge energy. The design provides this without needing to engage with screens or cognitively demanding tools. The design is meant to be worn indoors at rest.



Figure 7: a) Exhibition stand 06-06-2025 at Eindhoven University of Technology. b) Interactive prototype being worn.

5 Discussion and Future Work

This project was centered on exploring and developing a concept through an iterative design process, rather than on evaluating its clinical accuracy. The current interactive prototype serves as a demonstrator to communicate the concept. Due to ethical constraints, a formal user trial was not conducted with this

prototype. However, informal verbal feedback from users provided valuable insights for future development. Participants expressed a need to improve the comfort of the tight ropes, enhance the intuitiveness of putting on the garment, and further investigate the effectiveness of the haptic vibration cues in guiding breathing.

Additionally, several technical aspects must be addressed: the circuit needs to be made waterproof and modular for safety and washability. The signal accuracy, data richness, and clothing compatibility also require exploration. The algorithm for detecting dysfunctional breathing could be enhanced to be predictive and incorporate additional data. The breathing guidance should more fluently guide towards slow abdominal breathing.

Finally, while this project focused on long COVID, future versions could be tailored to specific subgroups within this population, such as those severely limited or those nearly recovered. At the same time, the core idea of breath guidance has broader relevance, from people with ME/CFS to those with general breathing difficulties and even to the healthy population, as everyone could potentially benefit from functional breathing

6 Conclusion

This project explored how interests in garment design, technology and inclusive design could merge into a supportive wearable for people with Long COVID, using a Reflective Transformative Design Process (RTDP). Long COVID remains a complex, sensitive and often misunderstood condition, characterized by diverse symptoms and needs. The shared interest from all participants in a supportive garment highlights its relevance.

A supportive garment, designed to create breath awareness, looks to help users recognize boundaries and relax without feeling restricted. Centered on user needs, it balances function and comfort by offering warmth, softness, and sensory safety with minimal invasiveness or effort. Breathing was used as a parameter for biofeedback and as a means to guide activity regulation and relaxation. Insights have been integrated into a functional, physical prototype.

The final interactive prototype lays a foundation for future development: a garment that supports breathing, promotes relaxation, and helps users with Long COVID find balance in their daily lives.

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